

Group 4 EDA Report — Verification Checklist

Document verified: [Group4_EDA.pdf](#) (final version, 16 pages) Verified against: [FIT5196S12026_assessment1.pdf](#) (spec), [A1_rubric.pdf](#) (rubric), [Group_group_number__EDA_template.pdf](#) (template)

A. Rubric Criterion 1 — Introduction and Conclusion (5%)

- [x] Report contains a dedicated Introduction section
- [x] Introduction states purpose, dataset scope, and methodology overview
- [x] Introduction identifies the 32,892 records and 18 attributes correctly
- [x] Report contains a dedicated Conclusion section (Section 5)
- [x] Conclusion summarises the five analytical phases
- [x] Conclusion restates the count of findings (10), insights (10), ML questions (5)
- [x] Conclusion's missing-data percentages match body (57.6% description, 55.7% City, 53.3% Country, 30.8% tags)
- [x] Conclusion's temporal scope claim matches body (Nov–Dec 2019, not 2007–2010)
- [x] Conclusion contains no information contradicting findings or insights
- [x] Conclusion avoids introducing new analytical claims not supported by findings

B. Rubric Criterion 2 — EDA Design (5%)

- [x] Design section is present (Section 2)
- [x] Design follows a logical, sequential pipeline
- [x] Each of the 5 phases is explicitly named and described (Structural Validation, Univariate, Bivariate, Multivariate, Insight Synthesis)
- [x] Phase descriptions specify which attributes/techniques are used per phase
- [x] Tools and libraries used are declared (Pandas, Matplotlib, Seaborn, NumPy, Jupyter)
- [x] Design is reproducible from description alone
- [x] Design maps to the 10 findings that follow

C. Rubric Criterion 3 — Findings (5%)

- [x] Exactly 10 findings provided (not 8, not 12 — count verified)
- [x] Each finding has a descriptive title statement, not a vague label

- [x] Each finding has supporting figure(s) with figure numbers
- [x] Each finding has prose justification with numeric evidence
- [x] Finding 1 numbers verified: 32,892 records, 18 attributes, 57.6% / 55.7% / 53.3% / 30.8% / 6.6% missingness
- [x] Finding 2 numbers verified: lat range 35.01–59.99°N, lon range 10.00–24.96°E, Sweden 2,195
- [x] Finding 3 numbers verified: germany 1,047 + deutschland 832 = 1,879; italia 1,238 + italy 758 = 1,996
- [x] Finding 4 numbers verified: 2,346 users, max 1,240 posts, median 2, mean 14.02
- [x] Finding 5 numbers verified: all 32,891 in 2019; Nov 28,882 (87.8%) + Dec 4,009 (12.2%); posting delay median 5, mean 60.8, max 915
- [x] Finding 5 title updated to remove misleading "seasonal peaks" phrasing
- [x] Finding 6 numbers verified: Stockholm 1,717 $\approx$ 2.56 $\times$ Washington DC 671
- [x] Finding 7 numbers verified: rates 0.498 / 0.489 / 0.349 (desc); 0.455 / 0.514 / 0.438 (loc); 0.643 / 0.669 / 0.650 (tag)
- [x] Finding 8 numbers verified: Pearson r = 0.170 (tags/delay), 0.132 (location/delay)
- [x] Finding 9 numbers verified: 40.0% both present, 49.0% both missing, 6.7% city-only missing, 4.3% country-only missing
- [x] Finding 10 numbers verified: top tags germany 2,437, deutschland 2,370, 2019 1,777, basketball 1,322, basket 1,241; title mean 28.2, desc mean 249.6, tag count mean 11.4
- [x] Findings are relevant to downstream insights and ML questions (cross-checked)

D. Rubric Criterion 4 — Insights and ML Questions (20%)

Insights

- [x] Exactly 10 insights provided (verified count)
- [x] Each insight has a bolded title claim followed by justification
- [x] Each insight is derived from specific findings (cross-references present)
- [x] Each insight contributes to research or business context, not merely descriptive
- [x] Insight 1 grounded in Finding 9 (geographic missingness)
- [x] Insight 2 grounded in Finding 3 (multilingual country naming)
- [x] Insight 3 grounded in Finding 4 (power-law distribution)
- [x] Insight 4 grounded in Finding 7 (metadata/activity relationship)
- [x] Insight 5 grounded in Finding 5 + Finding 10 (event window + basketball tags)
- [x] Insight 6 grounded in Finding 8 (tag/delay correlation)
- [x] Insight 7 grounded in Finding 6 (Stockholm concentration)
- [x] Insight 8 grounded in Finding 1 + Finding 7 (description missingness)
- [x] Insight 9 grounded in Finding 5 (weekend/midday peaks)
- [x] Insight 10 grounded in Finding 9 (city/country co-occurrence)
- [x] No insights are vague, speculative, or unsupported by data

ML Research Questions

- [x] Exactly 5 ML questions provided (verified count)
- [x] Each question is phrased as a specific, data-driven question
- [x] Each question has an explicit Justification subsection
- [x] Q1 specifies input features (lat/lon + tags), output (country), technique (Random Forest / GBT), **success metric (macro-F1 > 0.75, balanced recall)**
- [x] Q2 specifies input features (per-post metadata), output (activity tier), technique (LogReg / SVM / tree), **success metric (macro-F1 > 0.70, ROC-AUC > 0.80)**
- [x] Q3 specifies input features (tags, desc, timing, location, activity), output (posting delay), technique (Ridge / GBR), **success metric (MAE below naïve baseline, $R^2 > 0.30$)**
- [x] Q4 specifies input features (TF-IDF tags + coords + time), technique (K-Means / DBSCAN), **success metric (silhouette > 0.3, interpretable centroids)**
- [x] Q5 specifies input features (user + content level), output (desc present y/n), technique (LogReg / tree), **success metric (ROC-AUC > 0.75, F1 > 0.65, user-stratified split)**
- [x] Each question links explicitly to at least one finding/insight by number
- [x] Each question has clear business value articulated
- [x] Each question has research domain contribution articulated
- [x] None of the 5 questions require data not available in the dataset
- [x] Input features and output variables explained clearly for classification questions (spec requirement)

E. Rubric Criterion 5 — Report Structure and Presentation (5%)

- [x] Cover page present with report title and group identifier
- [x] Cover page has slots for all group members (Member 1–4 labels correct; no duplicated "Member 3")
- [x] Table of Contents present
- [x] TOC entries match actual section headings (re-verified after Word update)
- [x] TOC Finding 5 entry shows full heading text (not blank)
- [x] TOC Conclusion entry is a single line — no conclusion paragraphs bleeding in as TOC entries
- [x] Section numbering is consistent (1, 2, 3.1–3.10, 4.1, 4.2.1–4.2.5, 5)
- [x] All 18 figures are numbered sequentially (Figure 1 through Figure 18)
- [x] Every figure in the body is referenced in the prose
- [x] Figure captions present on each figure
- [x] References section present at end
- [x] References contain 9 entries (6 tooling + 3 methodology)
- [x] Methodology references added: Tukey (1977), Wickham (2014), Little & Rubin (2019)
- [x] Writing is scientifically toned throughout (no first-person, no casual register)
- [x] Heading styles applied consistently (Heading 1 / Heading 2 / Heading 3 hierarchy intact)

F. Spec Requirements — Structural and Procedural

- [x] Report filename matches spec: **Group4_EDA.pdf**
- [x] Report is a PDF (not .docx)
- [x] Report follows template structure (Intro → Design → Findings → Insights/ML → Conclusion → References)
- [x] Minimum 10 insights satisfied
- [x] Minimum 5 ML questions satisfied
- [x] Dataset record count (32,892) consistent everywhere in the report
- [x] Dataset attribute count (18) consistent everywhere in the report
- [x] Group number consistent throughout ("Group 4")

G. Internal Consistency Checks

- [x] Missing-data stats identical in Finding 1 body, Insight 8, and Conclusion
- [x] User activity thresholds (1–5 / 6–50 / 50+) identical in Finding 7, Insight 3, Insight 4, Q2 justification
- [x] Metadata completeness rates identical in Finding 7 and Insight 4
- [x] Temporal claim (Nov–Dec 2019) identical in Finding 5, Insight 5, Q3 justification, Conclusion
- [x] Stockholm count (1,717) and ratio (2.56×) identical in Finding 6 and Insight 7
- [x] Correlation values ($r = 0.170$, $r = 0.132$) identical in Finding 8 and Insight 6
- [x] No stat in the conclusion contradicts any stat in the body
- [x] No claim in any insight contradicts the finding it references
- [x] Basketball/basket event hypothesis consistently framed across Finding 10, Insight 5, Q4

H. Error and Typo Sweep

- [x] No placeholder "[Your Name]"-style text inside the body content
- [x] No stale "TODO" or bracketed draft notes
- [x] No broken figure references (every "Figure X" in prose exists)
- [x] No orphaned figure (every figure in the document is referenced in prose)
- [x] No duplicated section headings
- [x] No sections listed in TOC that are missing from body
- [x] No sections in body that are missing from TOC

I. Items Explicitly Excluded from This Verification

- [] **Group member names** — Bharath instructed to skip; user filling these in
- [] **Submission date** — user filling in
- [] **Abstract / executive summary** — user declined as optional
- [] **Normalisation of "posts" / "records" / "photos"** — user instructed to skip
- [] **.ipynb and .py file contents** — out of scope for report verification; user handles locally
- [] **AI declaration and AI record PDFs** — out of scope for report verification
- [] **Zip packaging** — out of scope; user handles at submission time

Verification outcome: All rubric-mapped items in Sections A through H are ticked. Report is cleared for submission once the items in Section I are actioned by the user at submission time.